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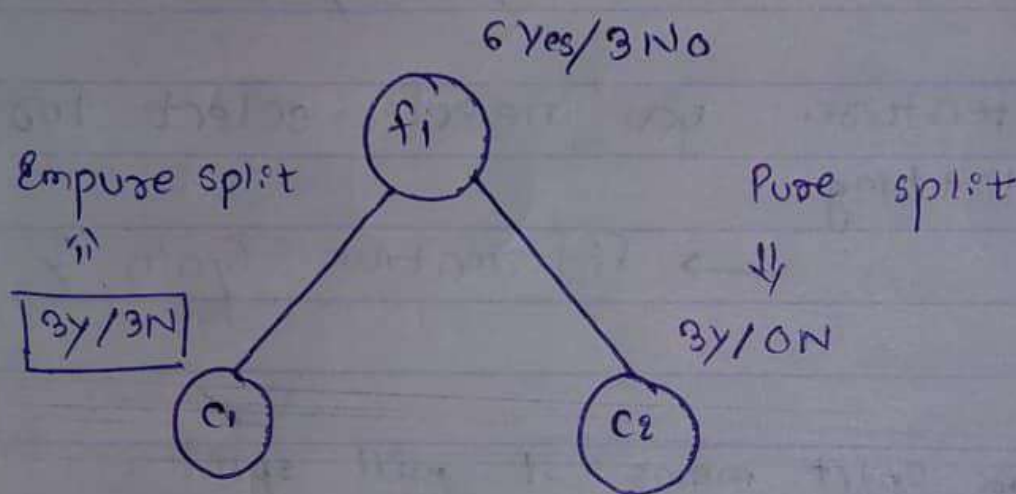
# Binary Classification

## 1) Entropy

## 2) Gini Impurity

$$H(S) = -p + \log_2 P + -p - \log_2 P -$$

$$G.I = 1 - \sum_{i=1}^n (P)^2$$



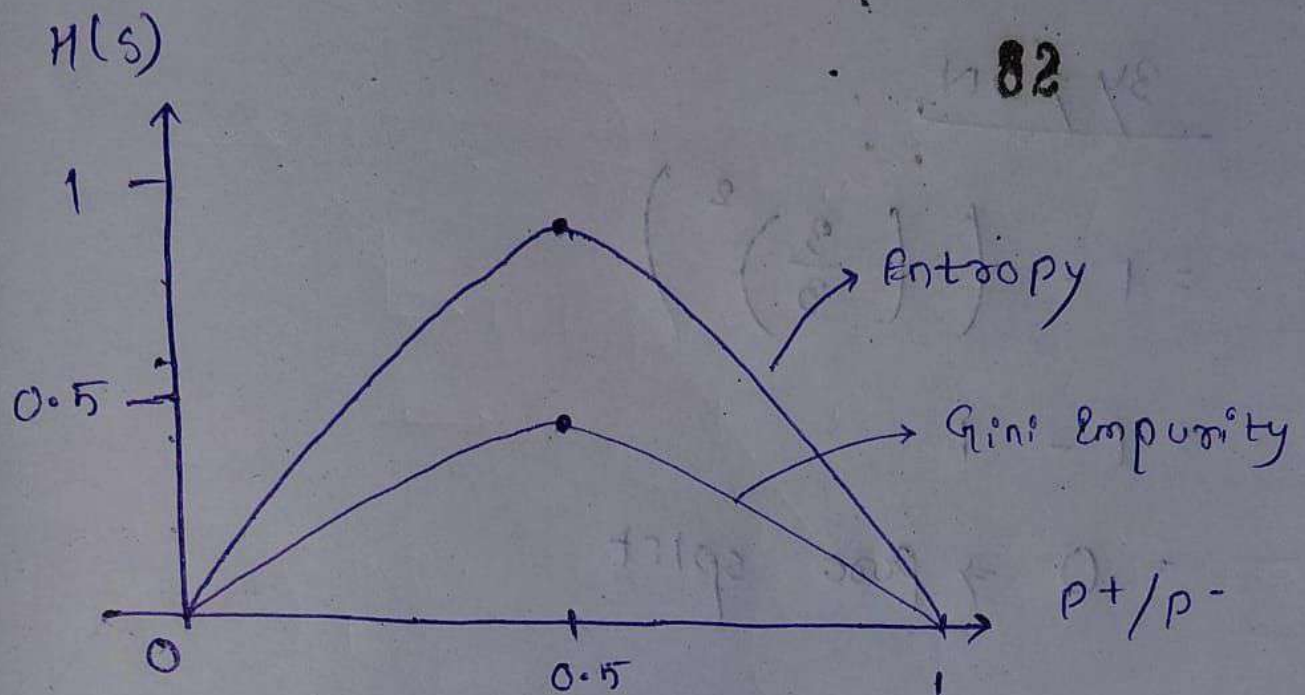
$$H(C_1) = -p + \log_2 P + -p - \log_2 P -$$

$$= -\frac{3}{6} \log_2 \frac{3}{6} - \frac{3}{6} \log_2 \frac{3}{6}$$

$$= 1 \Rightarrow \text{Impure split}$$

$$H(C_2) = -\frac{3}{3} \log_2 \frac{3}{3} - 0 \log 0$$

$$= -1 \log_2 1 \Rightarrow 0 \Rightarrow \text{Pure split}$$



② Gini Impurity

$$G.I = 1 - \sum_{i=1}^n (p_i)^2$$

$$= 1 - \left( (p^+)^2 + (p^-)^2 \right)$$

$$= 1 - \left( \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 \right)$$

0.5  $\Rightarrow$  Impure split

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$$= 1 - \left( \left( \frac{3}{3} \right)^2 \right)$$

$$= 1 - 1$$

= 0  $\Rightarrow$  pure split

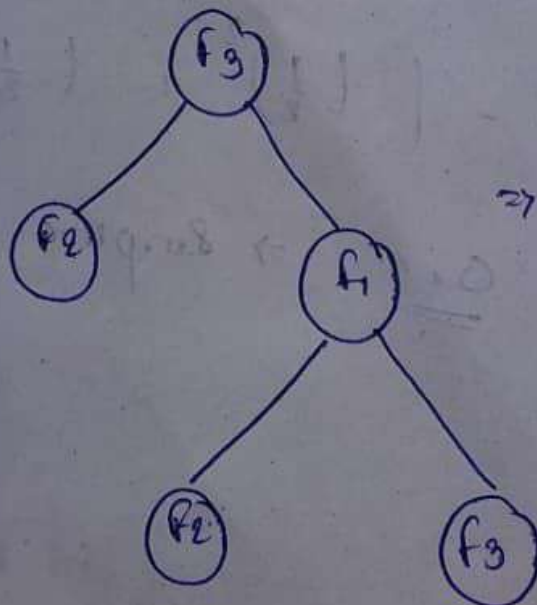
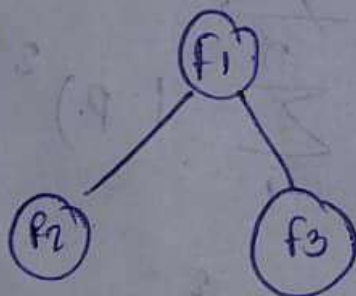
$f_1$

$f_2$

$f_3$

Decision  
split

Trees



$\Rightarrow$  Information  
Gain